Neuromatch Academy · Computational Neuroscience

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Does Biased Exposure Influence Choices Across Session Intervals?

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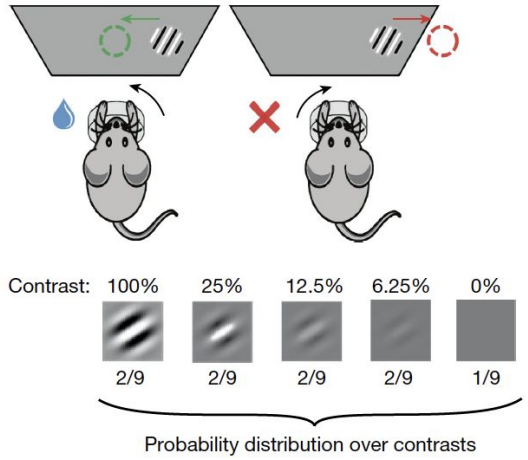
Pod: Each-Uisge Wormwood

Team: Horsing Around



Background · IBL Standardized Decision Making Task Structure

Perceptual decision making = current sensory evidence + prior expectations



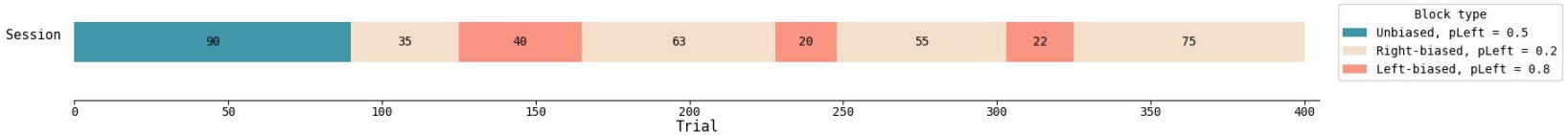
Trial Structure:

Head-fixed Mice turn a wheel to report the side of a visual grating for a reward

Fixed: first 90 trials always unbiased (50:50)

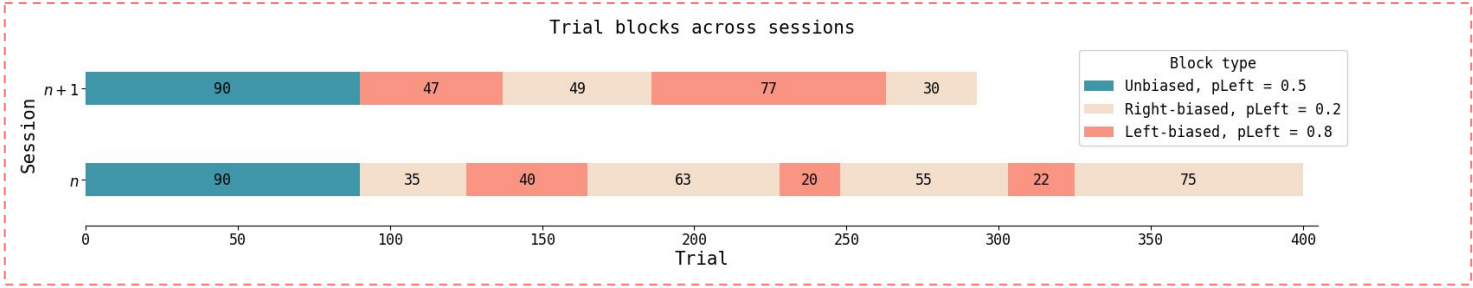
Patterned: after that, biased blocks *alternate* direction (80:20 ↔ 20:80)

Random: the *length* of each biased block is randomly drawn (20–100 trials, avg 50), and switches are unsignaled



Source: International Brain Laboratory et al., 2021

Mane Motivation · Intersession Carry-forward?



Imagine you're one of the mice.

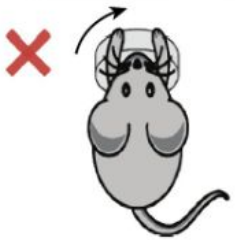
Yesterday's session ends after a biased block
Today begins with a 50:50 unbiased block

Does yesterday's session's bias carry forward into the unbiased trials of today?



Mouse choose left?





Mouse no choose left?

POV: you're a mouse and the wheel is your entire personality.

Previous Research & Gaps

Findling et al. (2025, Nature)	Examines how priors are represented within a session - **NOT** about persistence **across sessions**
Cochrane et al. (2023, npj Science of Learning)	Although it examines changes across sessions, these are quantified as improvements in decision accuracy, reaction time, and decision parameters - **NOT** persistence of the animal's **latent Bayesian prior**
Pinchuk-Yacobi et al. (2026, eLife)	Investigates how recent perceptual experience biases subsequent perception - **NOT** how **probabilistic beliefs are inferred.**
Zhang et al. (2026, Neuron)	Transfers information across sessions to enhance decoding accuracy, - **NOT** estimating **latent behavioral beliefs.**



Research Question & Hypothesis

Does the proportion of biased trials a mouse experiences during the previous session (n) influence choices in the next session's (n+1) unbiased block?

H_0: The proportion of left-biased trials in session n will not predict left-choice probability during the first 10 unbiased trials of session n + 1.

H_1 : A higher proportion of left-biased trials in session n will predict a higher probability of choosing left during the first 10 unbiased trials of session n + 1.

Methods · Sampling, Modelling & Analysis

Sample Details

- 214 consecutive session pairs (n and n+1)
- 26 fully trained mice
- Session gap: 12 - 24 hours

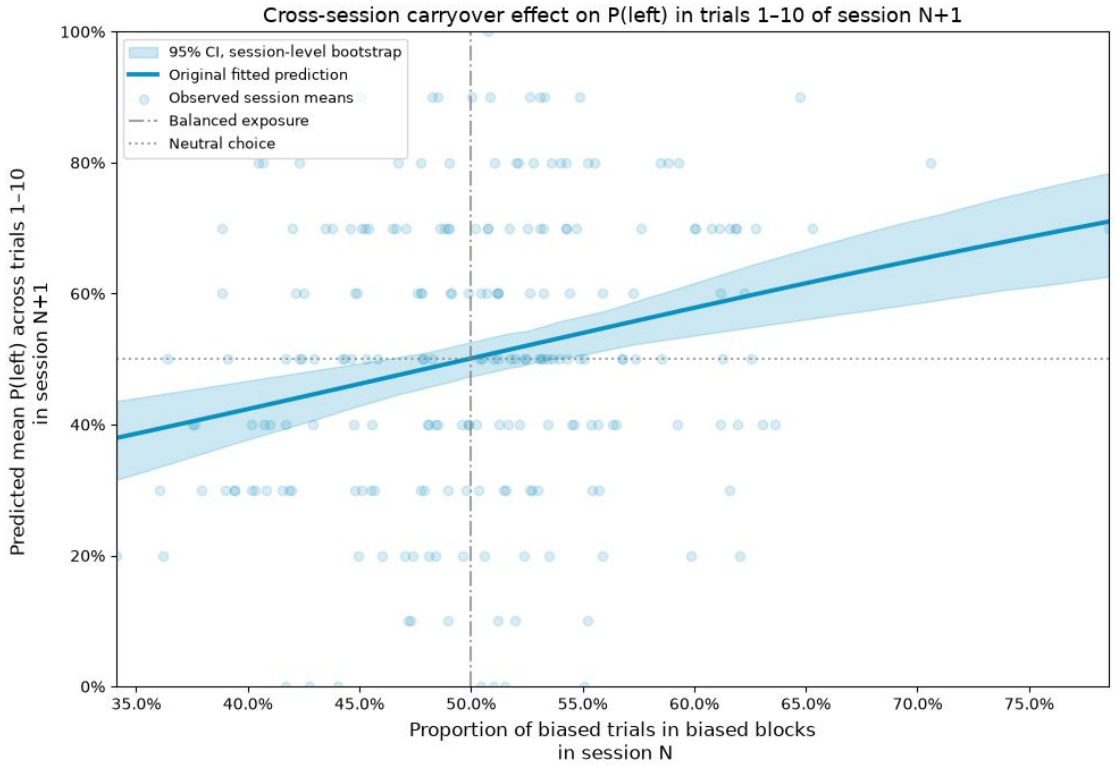
Model

- Bayesian binomial mixed-effects GLM with a logit link, fitted using variational Bayes
- Fixed effects: previous-session left-biased exposure, trial index
- Random intercepts: mouse and session, accounting for repeated measurements

Analysis

- Primary analysis: trials 1–10 in the unbiased block
- Zero-contrast analysis: only in the first 10 trials
- Within-mouse session-pair random permutation test (n=1000, p = 0.002)

Results · Cross Session Carryover in the First 10 Trials



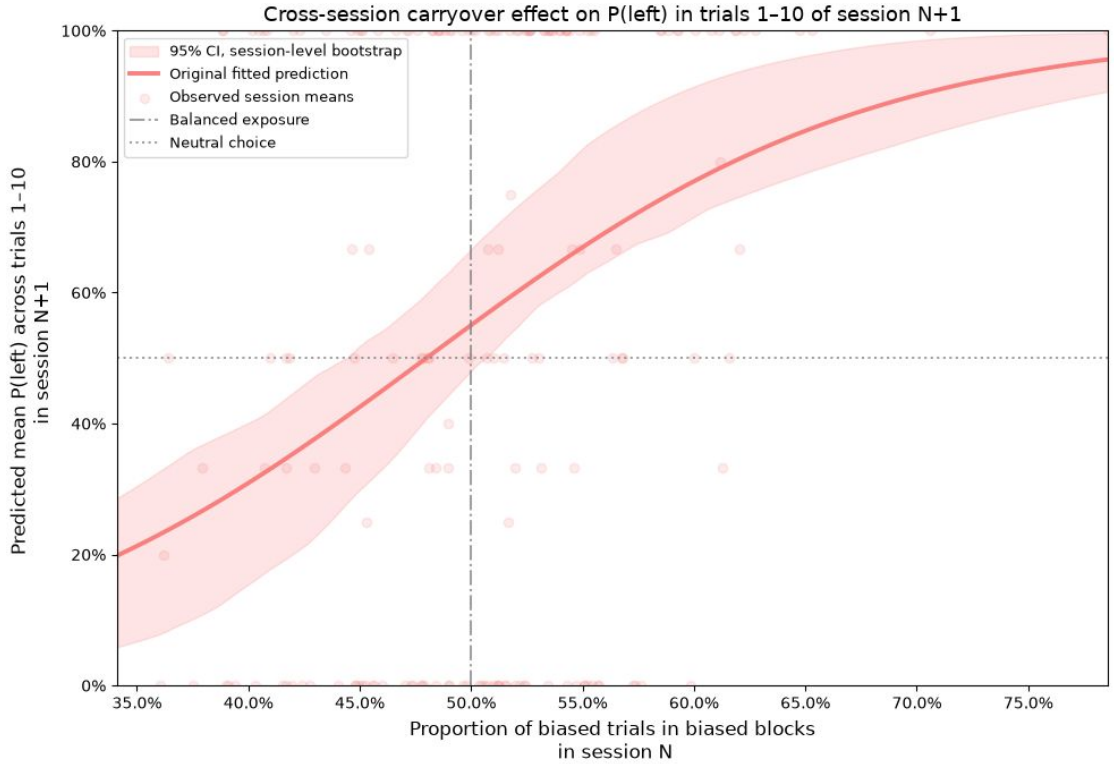
Evidence supports H_1, H_0 rejected

$\beta = 0.312$

95% Bootstrap CI = 0.194,0.455

Greater proportion of left biased trials in session n predicted a higher probability of choosing left in the first 10 unbiased trials of session n + 1

Results · Cross Session Carryover in Zero Contrast Trials (n = 385)



β = 1.01

95% Bootstrap CI = 0.714, 1.906

Greater left-biased exposure in session n predicted a higher probability of choosing left in session n+1, especially when the stimulus provided no directional information.

3.23

Slope ratio at 50% exposure

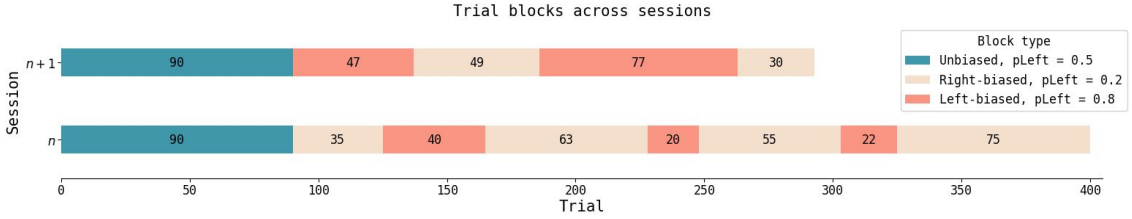
At 50% left-biased exposure, the zero-contrast relationship was estimated to be 3.23 times steeper than the relationship across all trials.

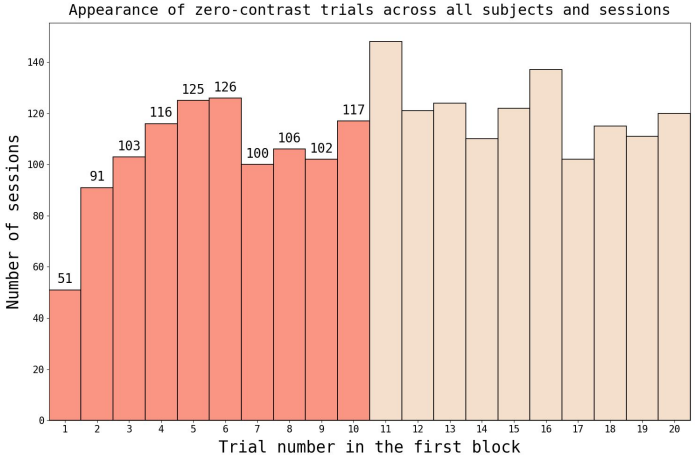

HOLD YOUR HORSES
imgflip.com


POV: What happens when you don't hold your horses


woah hold your horses

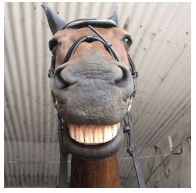
Study scope and Limitations



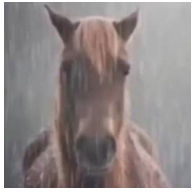


	Current constraint	Proposed next step
1	sessions separated by only 12–24 h	compare effects across time intervals
2	first 10 unbiased trials only	trace effect decay throughout the unbiased block
3	few and unevenly distributed zero-contrast trials	conduct a separate experiment focused on early zero-contrast trials
4	no tracking of effects across individual mice or between sexes	characterize distinct effect profiles across mice or between sexes

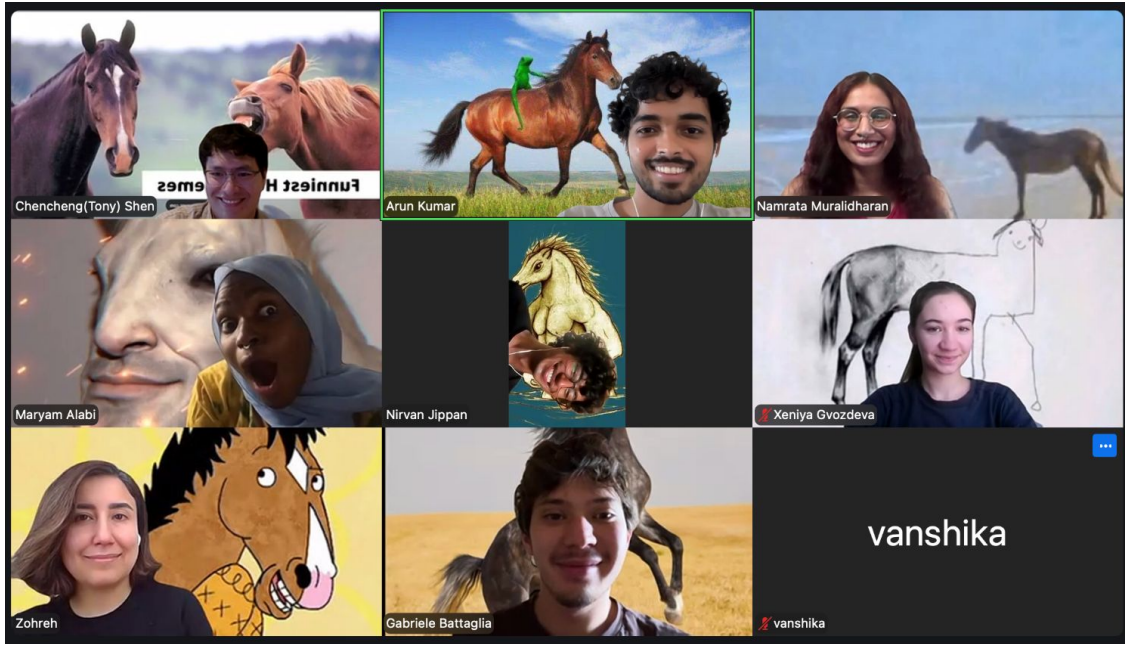
before limitations

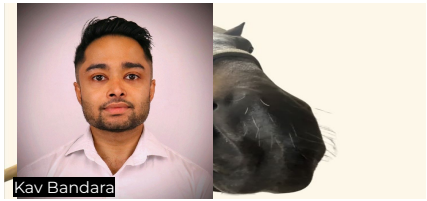


after limitations



Thank You · Questions?







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(and also) | All horse meme creators | Horses | Claude | ChatGPT | Each-Uisge

References

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